Ref. [40] — Lower Bounds for Non-Convex Stochastic Optimization

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1. The problem: is the slow rate of SGD unavoidable?

For non-convex stochastic learning, many algorithms prove that they eventually find a point where the gradient is small. But are their iteration rates loose analysis, or genuine limits of the information available? This paper constructs *lower bounds*: worst-case smooth optimization problems on which every method using a specified stochastic-gradient oracle needs many queries to find even an approximately stationary point. It does not ask for global or local minimizers.

$$\text{Target: } \|\nabla F(x)\| \le \epsilon, \qquad F : \mathbb{R}^d \longrightarrow \mathbb{R}.$$

2. Rebuild the standard SGD rate from first principles

Assume a smooth objective with Lipschitz gradient L and a lower bound F_*; we also have an unbiased stochastic gradient g_t satisfying

$$\mathbb{E}[g_t \mid x_t] = \nabla F(x_t), \quad \mathbb{E}[\|g_t - \nabla F(x_t)\|^2 \mid x_t] \le \sigma^2.$$

The smoothness (descent) inequality and SGD update $x_{t+1} = x_t - \eta g_t$ imply

$$\begin{aligned}\mathbb{E}_t F(x_{t+1}) &\le F(x_t) - \eta\|\nabla F(x_t)\|^2 + \frac{L\eta^2}{2}\mathbb{E}_t\|g_t\|^2 \\ &\le F(x_t) - \left(\eta - \frac{L\eta^2}{2}\right)\|\nabla F(x_t)\|^2 + \frac{L\eta^2\sigma^2}{2}.\end{aligned}$$

For $\eta \le 1/L$, telescope over $t = 0, \dots, T-1$ and define $\Delta = F(x_0) - F_*$:

$$\frac{1}{T}\sum_{t<T}\mathbb{E}\|\nabla F(x_t)\|^2 \le \frac{2\Delta}{\eta T} + L\eta\sigma^2.$$

Choosing η on the order of $T^{-1/2}$ makes the *expected squared gradient* decay as $O(T^{-1/2})$, so achieving it below ϵ^2 takes $T = O(\epsilon^{-4})$ with other parameters fixed. This is a teaching derivation of the familiar upper bound, not the paper's lower-bound proof.

3. The actual new contribution: two matching impossibility results

The authors prove that, for general L-smooth non-convex objectives with an unbiased stochastic gradient oracle having bounded variance, the order ϵ^{-4} cannot be uniformly improved: a suitable worst-case instance needs

$$\boxed{\Omega(\epsilon^{-4}) \text{ stochastic queries}}.$$

They also analyze a *more restrictive oracle*: the same random sample can give gradients that vary smoothly in mean square with the query location, typically expressed as

$$\mathbb{E}_z\|g(x,z) - g(y,z)\|^2 \le \widetilde{L}^2\|x - y\|^2.$$

In this class they establish a sharper lower bound of $\Omega(\epsilon^{-3})$. This agrees with the best recursive variance-reduced gradient methods of the corresponding class (such as SPIDER-type estimators). Smoothness of the *expected* function alone is weaker than mean-squared smoothness of the individual stochastic oracle.

> **Main result.** Under bounded-variance stochastic gradient information, the classical ϵ^{-4} worst-case rate is minimax optimal. With mean-squared smoothness of the oracle, the correct frontier improves to ϵ^{-3}. The result is about *first-order stationarity*, not escaping every saddle.

4. Why constructing lower bounds is different from convergence analysis

An upper bound describes an algorithm and proves *it succeeds* within a specified number of queries on all admissible functions. A lower bound must show that *every possible oracle algorithm* faces a hard admissible function; one counterexample for ordinary SGD would not suffice. The paper uses high-dimensional, carefully hidden information and noisy oracle constructions. A picture of a long narrow valley can build intuition, but it is *not* a substitute for that information-theoretic argument.

5. Assumptions, limits, connection to our survey

The guarantee is minimax and asymptotic in the stationarity tolerance, with variance, smoothness, initial gap, and dimension relevant to complete statements. It does not predict convergence time for a particular neural net, incorporate GPU parallelism, or require the hard instance to resemble practical ReLU losses. It does not prove second-order stationarity or any particular saddle escape mechanism. This is core **RQ3** material for the oracle complexity comparison and **RQ4** for explaining why strong performance claims depend on the oracle model.

Important distinction. Do not conflate the $\Omega(\epsilon^{-3})$ lower bound under *mean-squared oracle smoothness* with an ϵ^{-3} bound under mere bounded variance. They are different assumptions.

6. Our notes

This paper is the reason we should not call SGD's ϵ^{-4} rate "outdated" just because SPIDER can get ϵ^{-3}. SPIDER relies on a stronger way of comparing noisy gradients at nearby points. Under the weaker stochastic oracle, the slower rate is actually optimal in the worst case. It also keeps our table honest: small gradient does not mean we have escaped a saddle. **Original source:** https://doi.org/10.1007/s10107-022-01822-7. **Source check:** full author manuscript accessible via the NSF Public Access Repository.